Form A

1. (a) and (c)

2. (c) only

4. (a) 99.7%

   (b) $np \geq 10$ and $n(1-p) \geq 10$.

   (c) It is reasonable so long as the ratio: $\dfrac{\text{population size}}{\text{Sample size}} \geq 20$.

5. (e) gf-qq(~differences)

   (f) The dots do not perfectly align with a line, but they mostly do, making it appear that the population of differences

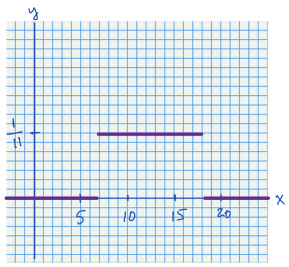
   glaucoma − healthy

   is somewhat like a normal distribution.

   (g) The 90% CI for μ_{diff} is

   $$-2.5 \pm (1.833113)\frac{2.5927}{\sqrt{10}} \quad \text{or} \quad (-4.003, -0.997)$$

6. (a)



   (b) qunif(0.36, 7, 18)

7. (a) The sample proportion has $SE_{\hat{p}} = \sqrt{\dfrac{p(1-p)}{n}}$, and a test (z) statistic:

   $$\hat{p} = \frac{46}{92} = 0.5 \quad \Rightarrow \quad z = \frac{0.5 - 0.41}{\sqrt{\dfrac{(0.41)(0.59)}{92}}} \doteq 1.755$$

   (b) 2 * (1 - pnorm(1.755)) (2-sided)

   (c) Letter (iii)

   (d) The result is not statistically significant at the 3% level. We fail to reject that $p = 0.41$.

8. (a) The samples are "independent" and "random," which we need. Though sample sizes (19 and 23) are close to 30, we have no knowledge of the population to assure us that $\bar{x}_1, \bar{x}_2$ are normal, and thereby allowing small sample sizes.

(b) $qt(0.96, df = 37.5976)$

(c) Take μ_1 = mean hemoglobin level for breast-fed infants

μ_2 = mean hemoglobin level for formula-fed infants

$$\bar{x}_1 - \bar{x}_2 \pm t_c \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = (13.3 - 12.4) \pm (1.799)\sqrt{\frac{1.7^2}{23} + \frac{1.8^2}{19}}$$

or $(-0.791, 1.879)$